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MODULE CODE: CET138

MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

SUBMISSION DATE AND TIME: Fri, 19 Jun 2026

ASSIGNMENT: 1

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CET138 Full Stack Development

JAVASCRIPT DOCUMENTATION

Details	
Name	Binod Katuwal
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Module Code	CET138
Module Title	Full Stack Development
Module Leader	David Grey
Submission Date	Fri, 19 Jun 2026
Assignment	Assignment 1 – portfolio (30%)
Project	Forever Moments – JAVASCRIPT Demo
Live Demo	https://binodkatuwal.github.io/Demos/javascriptdemo.html

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1. What is JavaScript?

JavaScript (JS) is a high-level, interpreted programming language that adds interactivity and dynamic behavior to web pages. HTML gives the page structure, CSS gives it style, and JavaScript is used to change the content of the page in real time, depending on the user's actions, such as clicking, scrolling, loading the page, etc.

DOM manipulation: Select and modify HTML elements, event handling: run code on user interaction, control flow: use conditionals, loops and timers for dynamic behavior. The following live demo ("Forever Moments") combines three of the classic interactive widgets from JavaScript into a single page.

2. Live Demo

[View Live Demo: Forever Moments – JavaScript Demo](#)

The page demonstrates three separate JavaScript-driven features: a dark/light theme switcher, ("Flip Color"), animated counters, and an interactive FAQ accordion. Each is explained below with the underlying JavaScript concepts.

3. Feature 1: Theme / Color Switcher

A button labelled "Flip Color" toggles the page between a dark theme and a light theme. This is achieved by toggling a CSS class on the <body> element — JavaScript adds or removes the class, and CSS handles the actual color change.

```
const flipBtn = document.querySelector('.flip-color-btn');

flipBtn.addEventListener('click', () => {
  document.body.classList.toggle('light-theme');

  // Optional: change button text based on current theme
  flipBtn.textContent = document.body.classList.contains('light-theme')
    ? 'Dark Ink'
    : 'Flip Color';
});

/* CSS */
body { background: #0E0B07; color: #fff; transition: background 0.4s, color 0.4s; }
body.light-theme { background: #FFF8F0; color: #2b2b2b; }
```

- `document.querySelector()` — selects the button element from the DOM
- `addEventListener('click', ...)` — runs a function when the button is clicked
- `classList.toggle()` — adds the class if absent, removes it if present
- CSS transition — makes the color change animate smoothly rather than snap instantly

4. Feature 2: Animated Counters

The 'Animated Counters' section displays statistics (Weddings, Happy Clients %, Years Active, Partners) that count up from 0 to their target value. This uses `setInterval` (or `requestAnimationFrame`) to repeatedly increment a number and update the DOM until the target is reached.

```
counters.forEach(counter => {
  const target = parseInt(counter.dataset.target, 10); // e.g. data-target="250"
  let current = 0;
  const step = Math.ceil(target / 100); // controls speed

  Windsurf: Refactor | Explain | Generate JSDoc | X
  const update = () => {
    current += step;
    if (current >= target) {
      counter.textContent = target; // stop exactly on target
    } else {
      counter.textContent = current;
      requestAnimationFrame(update); // smooth animation loop
    }
  };
  update();
});
```

- `querySelectorAll()` — selects all counter elements at once
- `dataset.target` — reads a custom `data-target="..."` HTML attribute
- `requestAnimationFrame()` — smoothly repeats the update for animation
- `textContent` — updates the visible number inside the element

5. Feature 3: FAQ Accordion

The FAQ section lists questions (e.g. 'What is included in the Classic Package?'). Clicking a question expands or collapses its answer. Only the clicked item's content is toggled, and JavaScript can optionally close other open items so only one answer is visible at a time.

```
const faqItems = document.querySelectorAll('.faq-item');

faqItems.forEach(item => {
  const question = item.querySelector('.faq-question');

  question.addEventListener('click', () => {
    const isOpen = item.classList.contains('open');

    // Close all items first (accordion behaviour)
    faqItems.forEach(i => i.classList.remove('open'));

    // Re-open this one only if it was previously closed
    if (!isOpen) {
      item.classList.add('open');
    }
  });
});

/* CSS */
/*
.faq-answer { max-height: 0; overflow: hidden; transition: max-height 0.3s ease; }
.faq-item.open .faq-answer { max-height: 200px; }
.faq-item.open .faq-question::after { content: '-'; }
*/
```

- `forEach()` — loops through every FAQ item to attach a click listener
- `classList.contains()` — checks whether an item is currently open
- `classList.add()` / `remove()` — controls the open/closed state
- CSS `max-height` transition — animates the expand/collapse smoothly

6. Summary

The "Forever Moments" demo brings together three classic, widely-used JavaScript UI patterns in a single page, demonstrating a thorough range of front-end JavaScript skills:

- DOM selection with `querySelector()` and `querySelectorAll()`
- Event handling with `addEventListener()`
- Dynamic class manipulation with `classList` (`toggle`, `add`, `remove`, `contains`)
- Animation logic using `requestAnimationFrame()` for smooth counters
- Reading custom data attributes (`data-target`) from the DOM
- Combining JavaScript logic with CSS transitions for smooth visual effects